

Computational hwk #1

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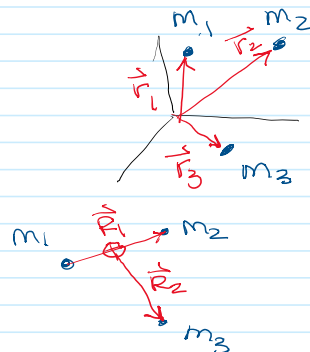
3/20/26

1) given: 3BP EOM in Jacobi coord

$$m_1 + m_2 + m_3 = 1 \quad G = 1$$

input: masses, $\vec{R}_0$, $\vec{V}_0$

output: time series of $\vec{r}$ & $\vec{v}$



$$\vec{R}_1 = \vec{r}_{12} = \vec{r}_2 - \vec{r}_1$$

$$\vec{R}_2 = \frac{1}{m_1 + m_2} [m_1(\vec{r}_3 - \vec{r}_1) + m_2(\vec{r}_3 - \vec{r}_2)]$$

$$\vec{R}_N = \vec{r}_N$$

$$\vec{R}_{N-1} = \frac{1}{\sum_{i=1}^{N-1} m_i} \sum_{i=1}^{N-1} m_i \vec{r}_{iN}$$

$$\vec{r}_{12} = \vec{R}_1$$

$$\vec{r}_{23} = \vec{R}_2 - \frac{m_1}{m_1 + m_2} \vec{R}_1$$

$$\vec{r}_{13} = \vec{R}_2 - \frac{m_2}{m_1 + m_2} \vec{R}_1$$

$$M_i = \sum_{j=1}^N m_j$$

$$T = \sum_{i=1}^{N-1} M_{i+1} \frac{M_{i+1} m_i}{M_i} \dot{\vec{R}}_i \cdot \dot{\vec{R}}_i + \frac{1}{2} M_N \dot{\vec{R}}_N \cdot \dot{\vec{R}}_N$$

$$H = \sum_{i=1}^{N-1} \frac{M_{i+1} m_i}{M_i} \dot{\vec{R}}_i \times \dot{\vec{R}}_i + M_N \dot{\vec{R}}_N \times \dot{\vec{R}}_N$$

$$I_p = \sum_{i=1}^{N-1} \frac{M_{i+1} m_i}{M_i} R_i^2 + M_N R_N^2$$

$$\vec{y} = \begin{bmatrix} \vec{R} \\ \dot{\vec{R}} \end{bmatrix} \quad \frac{d\vec{y}}{dt} = \dot{\vec{y}} = \begin{bmatrix} \dot{\vec{R}} \\ \ddot{\vec{R}} \end{bmatrix}$$

$$U = - \sum_{i=1}^{N-1} \sum_{j=i+1}^N \frac{G m_i m_j}{|\vec{r}_{ij}|}$$

Lagrangian Eqs $\Rightarrow$ EOM

$$\frac{\partial \vec{r}_i}{\partial \vec{R}_j} = \begin{cases} 0 & j \geq i \\ 1 & j = i-1 \\ \frac{m_{j+1}}{m_{j+1} + 1} & j \leq i-2 \end{cases}$$

$$\frac{\partial U}{\partial \vec{R}_k} = -G \sum_{l=1}^{N-1} \sum_{j=l+1}^N \frac{m_l m_j}{r_{lj}^3} \vec{r}_{lj} \cdot \frac{\partial \vec{r}_{lj}}{\partial \vec{R}_k}$$

$$L = T - U$$

$$\frac{\partial L}{\partial \vec{R}_j} = \frac{M_{j+1} m_j}{M_j} \dot{\vec{R}}_j$$

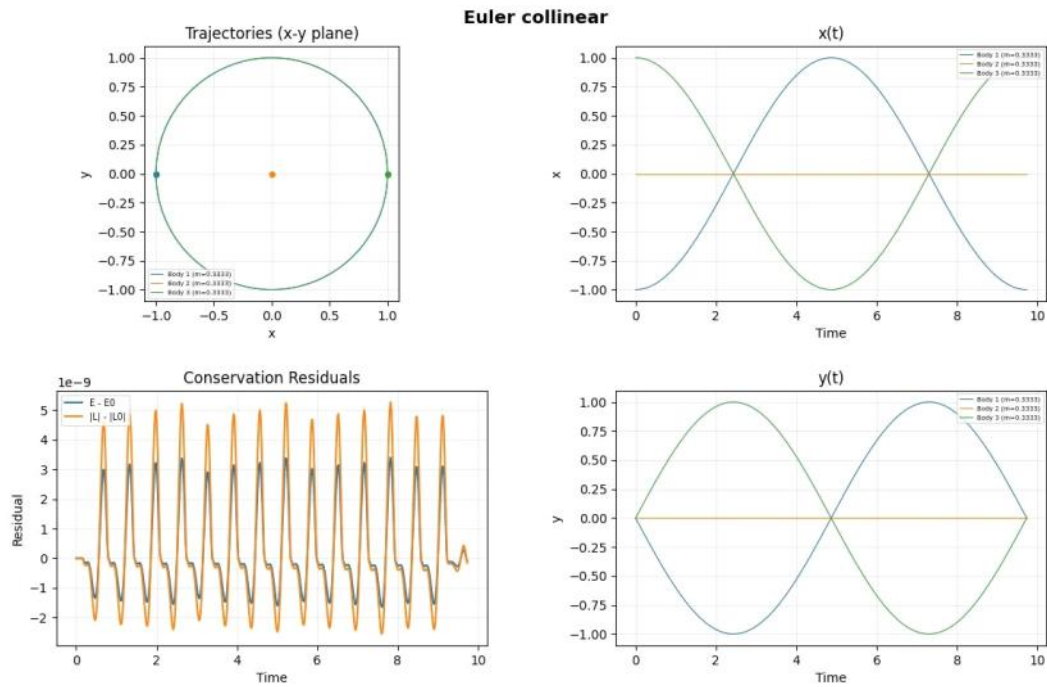
$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\vec{R}}_j} \right) = \frac{M_{j+1} m_j}{M_j} \ddot{\vec{R}}_j$$

1a) rel tol = 1×10^{-8}

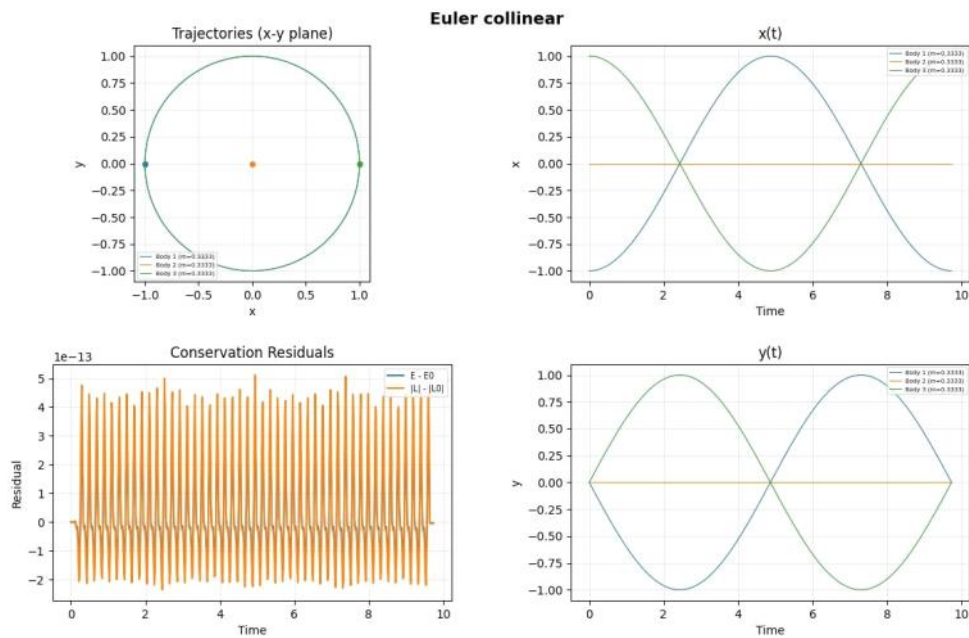
abs tol = 1×10^{-10}

(using Euler
const mass conf(q))

1a) $\text{rel tol} = 1 \times 10^{-8}$ (using Euler
 $\text{abs tol} = 1 \times 10^{-10}$ const mass config)

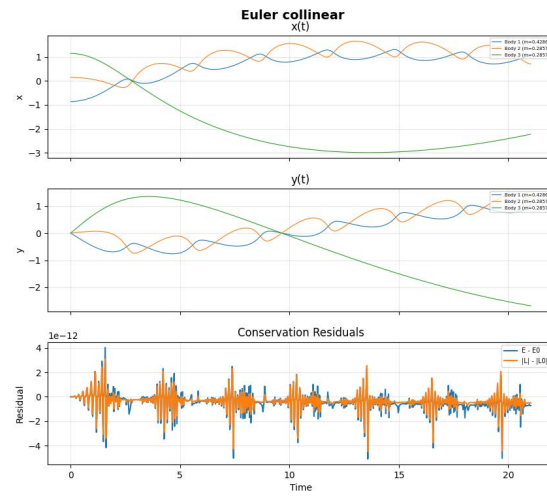
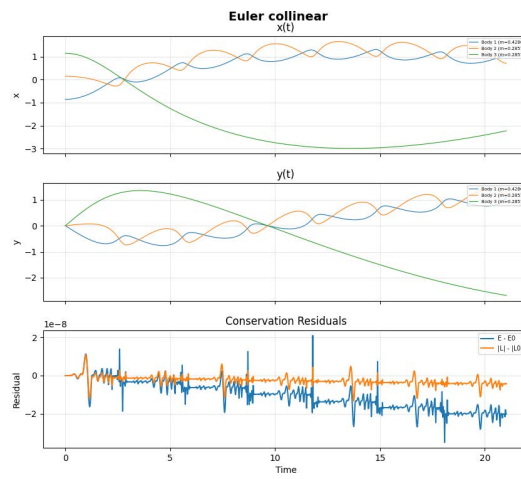
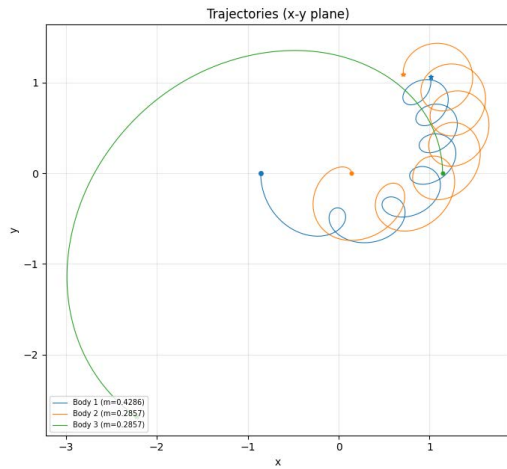


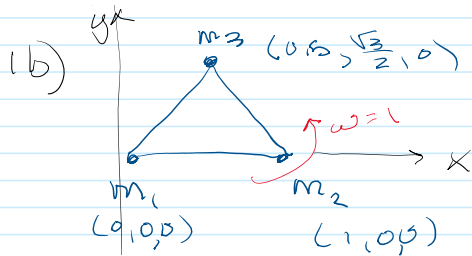
$\text{rel tol} = 1 \times 10^{-12}$
 $\text{abs tol} = 1 \times 10^{-14}$



Euler collinear w/ unequal masses

Euler collinear



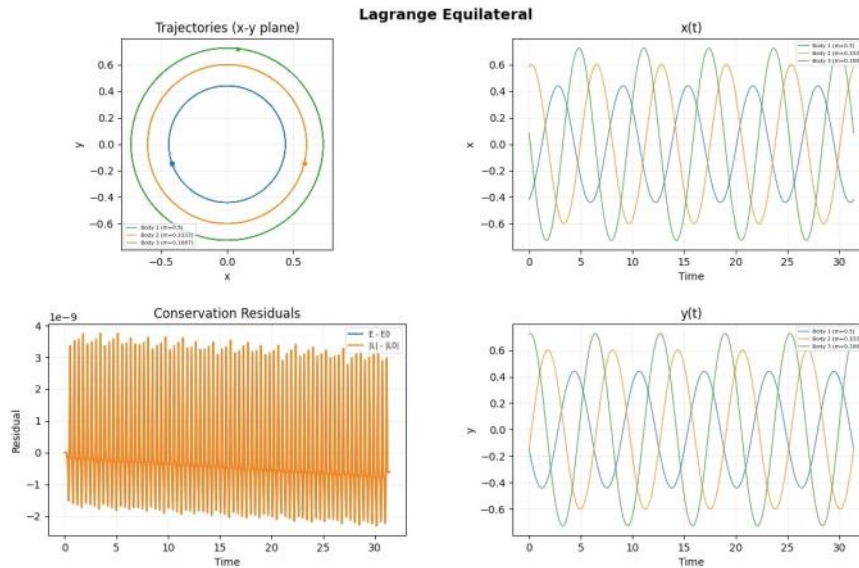


$$m_1 = \frac{1}{2} \quad m_2 = \frac{1}{3} \quad m_3 = \frac{1}{6}$$

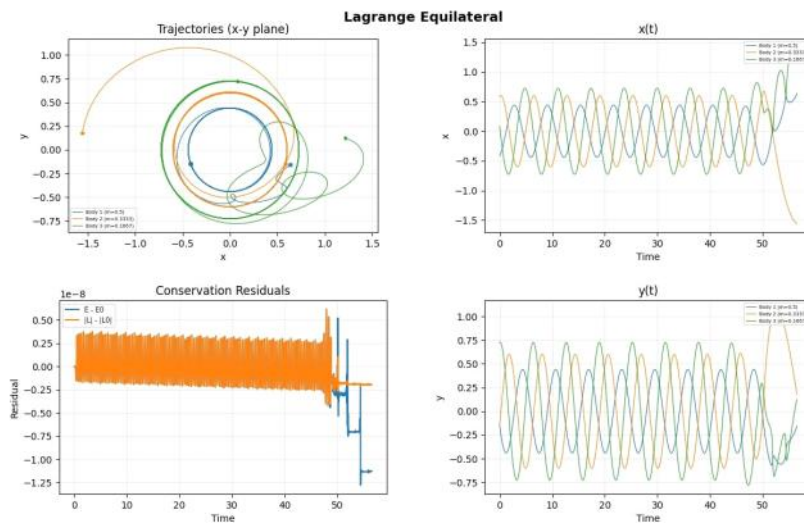
$$\omega = 1 \quad \vec{v} = \omega \vec{r}$$

$$\text{period} = 2\pi / \omega$$

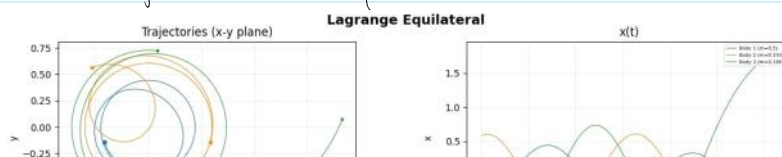
The orbit stays stable through about 8 periods

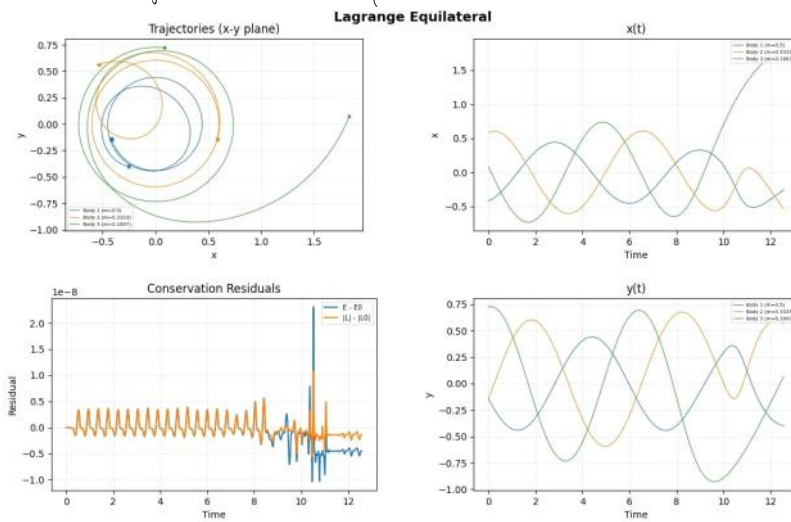


During the 9th period, it becomes quite chaotic

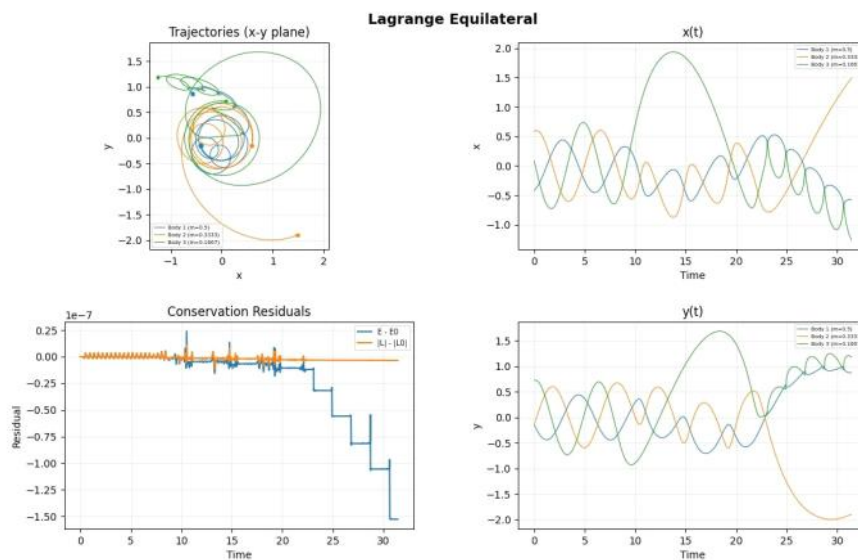


Adding a small perturbation to a single body (body 2) results in a large difference in each body's trajectory, this after 2 periods





This is after 5 periods:



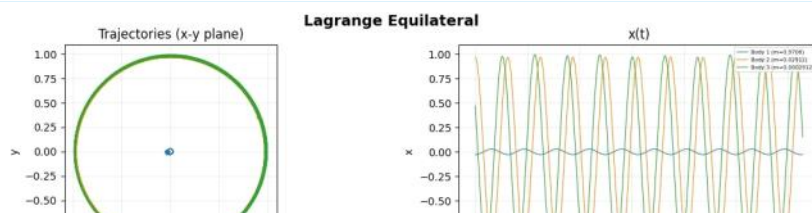
(c) repeating (b) but with masses that meet the Routh stability criterion

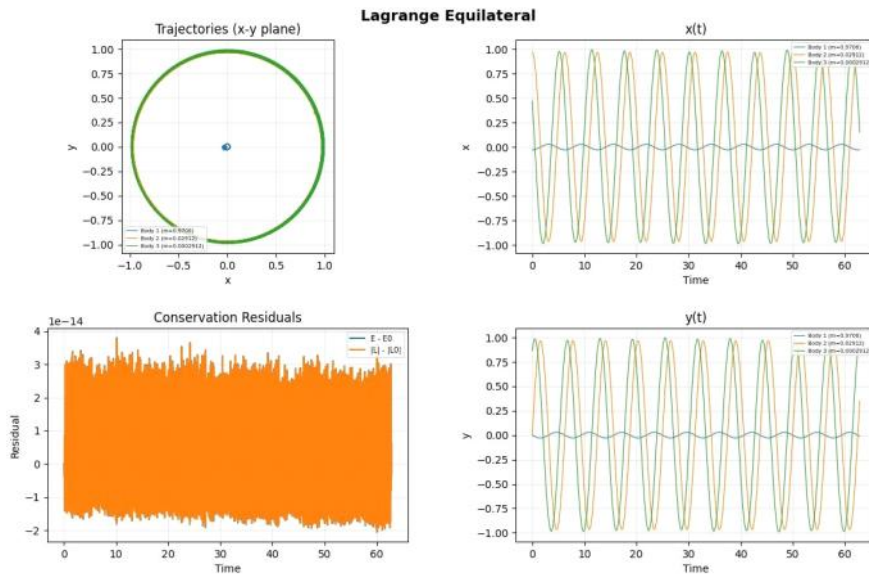
$$m_1 = \frac{1}{m_1 + m_2 + m_3}$$

$$m_2 = \frac{0.03}{m_1 + m_2 + m_3}$$

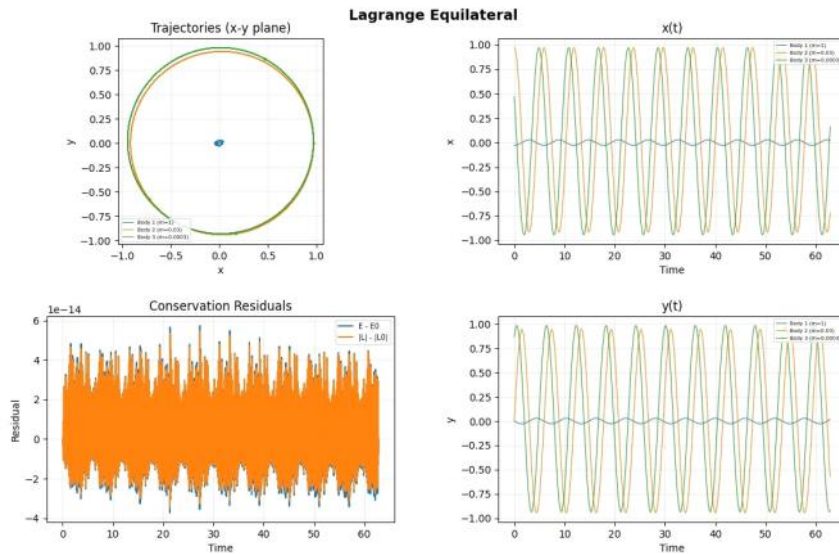
$$m_3 = \frac{0.0003}{m_1 + m_2 + m_3}$$

They are stable & stay in equilateral triangle config



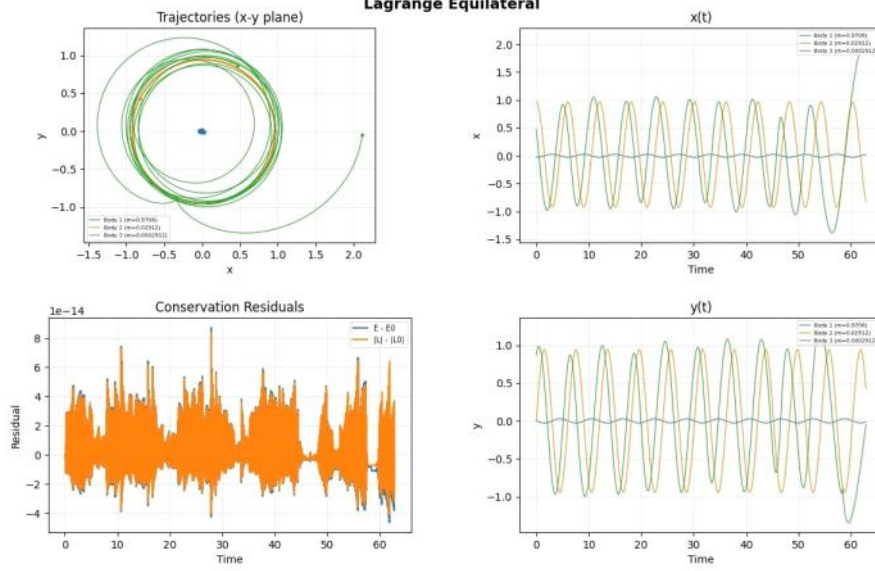


Adding 0.001 to r_2 results in small deviations but system is still stable & in equilateral triangle config.



The system stays stable thru the 1st 10 orbits until I perturb m_2 by 0.007. Then the system config moves away from triangular config.

Lagrange Equilateral



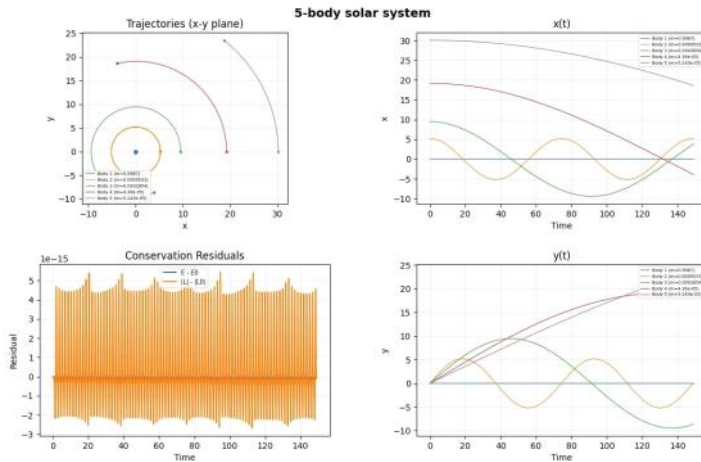
Problem 2

2a) I used the outer planets to verify
SBP - Sun, Jupiter, Saturn, Uranus, Neptune

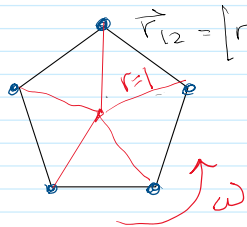
$$m_{\text{Sun}} = 1 \quad m_J = 9.54 \times 10^{-4} \quad m_S = 2.86 \times 10^{-4} \quad m_U = 4.37 \times 10^{-5} \quad m_N = 5.15 \times 10^{-5}$$

$$V = m_{\text{Sun}}/r \quad r_J = 5.2 \text{ Au} \quad r_S = 9.5 \text{ Au} \quad r_U = 19.2 \text{ Au} \quad r_N = 30.1 \text{ Au}$$

Energy & momentum are conserved thru
2 Jupiter orbits.



2b)

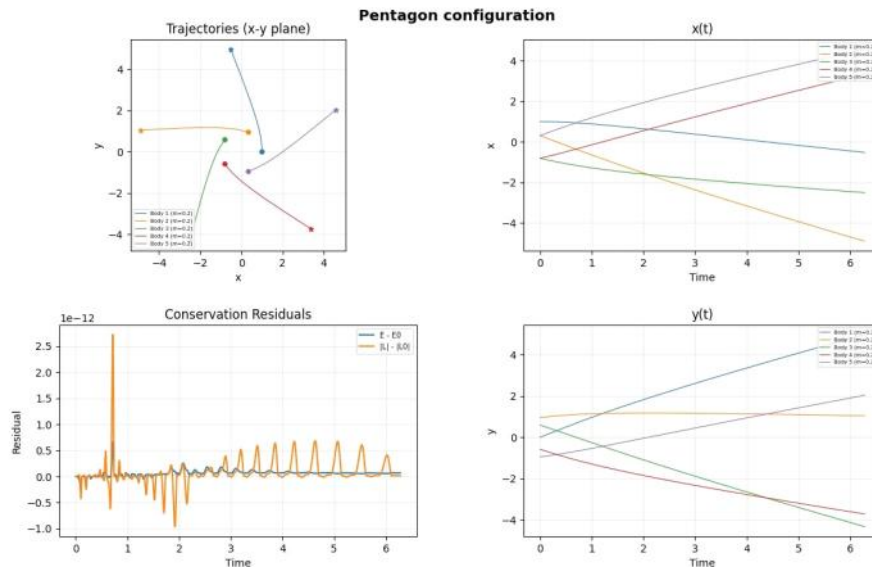


$$\vec{r}_i = [r \cos(2\pi i/5), r \sin(2\pi i/5), 0]$$

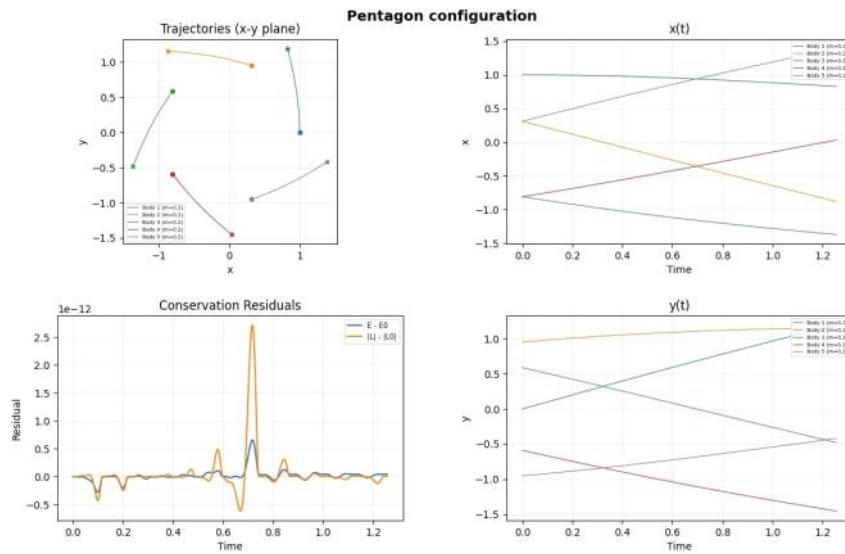
$$m_i = \frac{1}{5} \quad \omega = \frac{GM}{r^3}$$

$$v_i = \omega \vec{r}_i$$

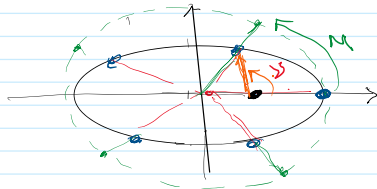
In this configuration the bodies fly
away immediately.
Here is one period:



Here is 0.2 period



2c) 5 bodies in eccentric orbit



$$a = 1 \quad e = 0.1 \quad \text{Period} = 2\pi \sqrt{\frac{a^3}{\mu}}$$

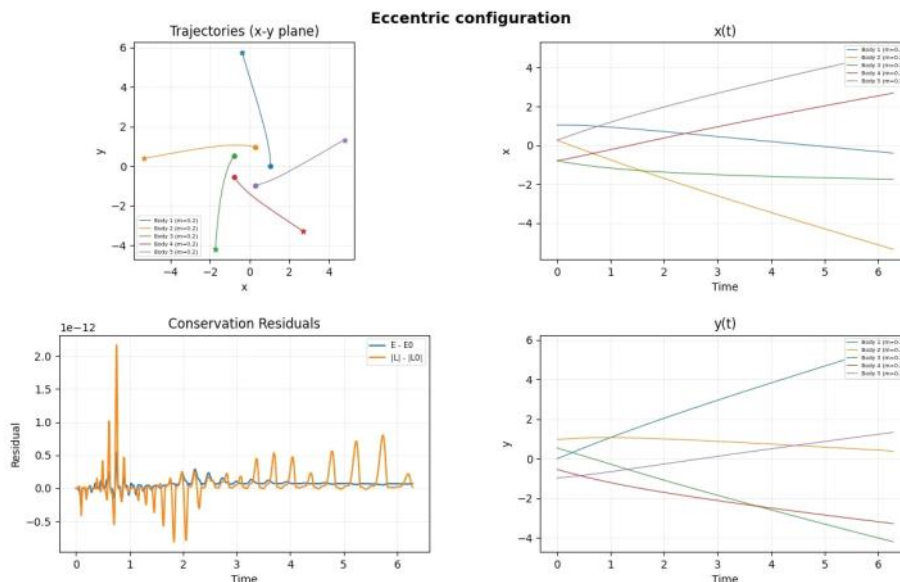
$$h^2 = \mu a (1 - e^2)$$

M = mean anomaly

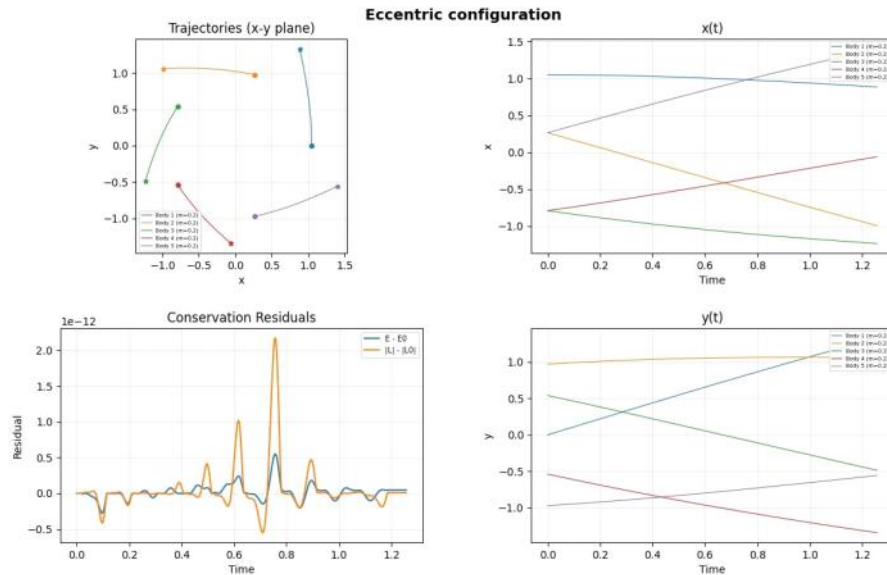
$$f = E - E_0 \sin(e) - M$$

$$\tan \frac{y}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2}$$

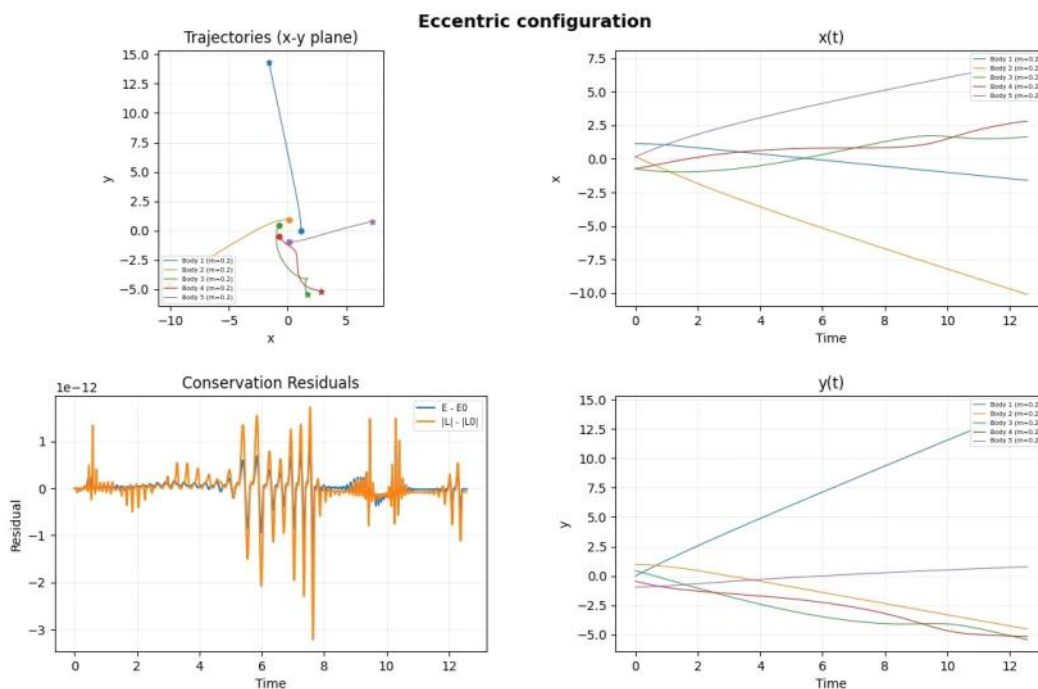
The system is still unstable & the masses fly away from each other immediately but since there the masses have non uniform velocity, the orbit departures are "shallower" since they are on an ellipse the velocity components are slightly different.



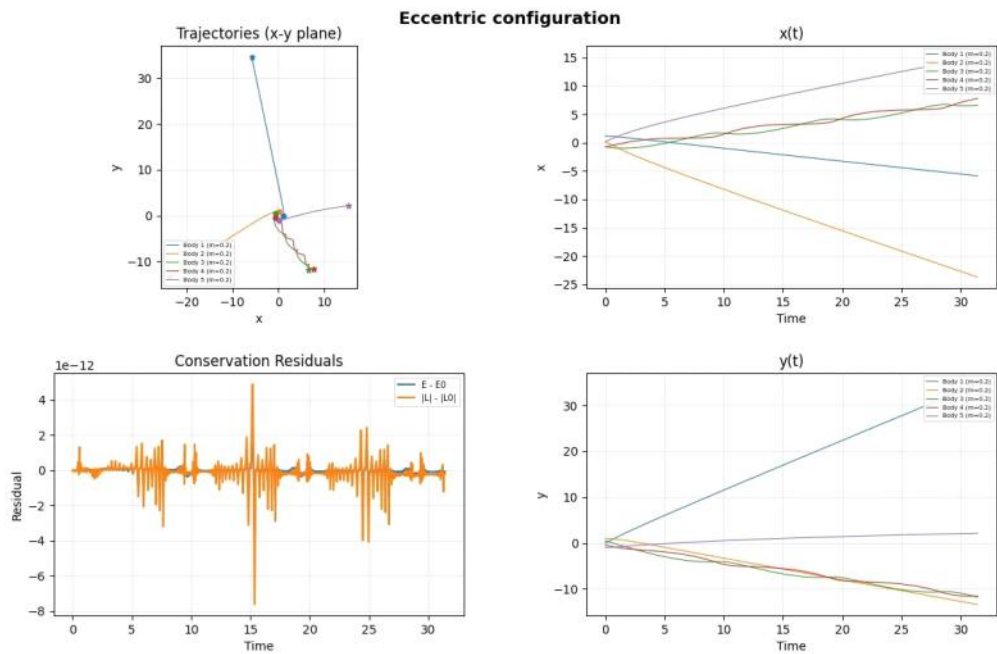
This 0.2 period



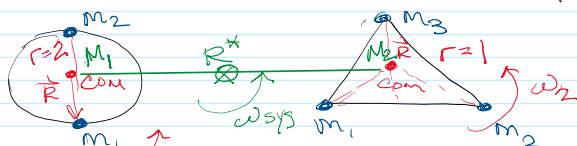
As the eccentricity is increased, the body @ periaapsis leaves the system fastest & travels farther a way than the other bodies.
 This is $e=0.3$ over the course of 2 periods. The bodies near apoapsis continue to impact the others orbit because their velocity is slowest.



After 5 periods they are still bound to each other.



2d) 2 body & 3 body system
orbit each other & then fly by



$$\vec{R} = \frac{1}{\sqrt{3}}$$

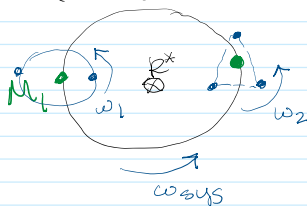
$$V = \sqrt{\frac{M_1 + M_2}{r_{12}}}$$

$$r_{12} = 2 \quad V = \sqrt{\frac{2}{5/2}} = \sqrt{\frac{4}{5}}$$

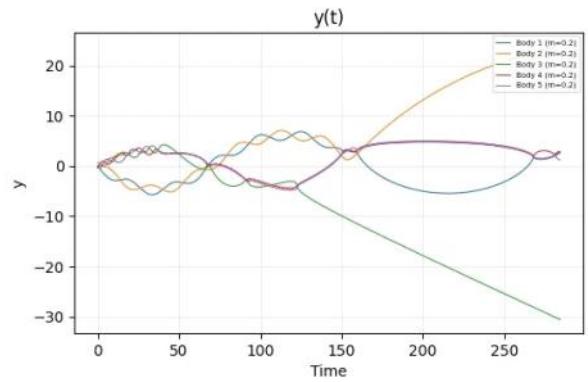
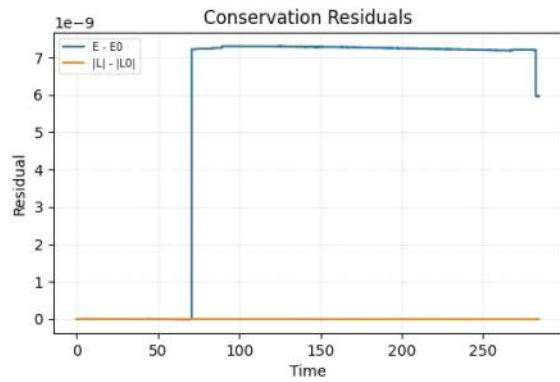
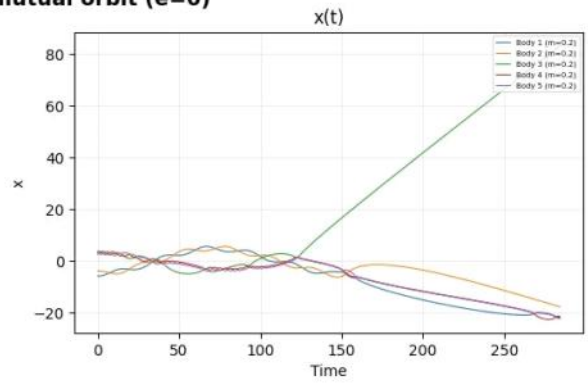
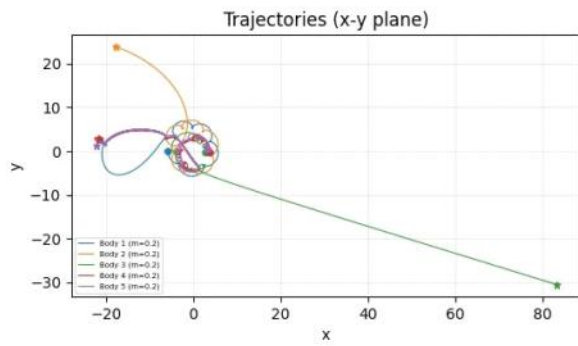
$$r_{12} = r_{23} = r_{13} = 1 \quad V = \sqrt{\frac{3}{5}}$$

start w/ all masses equal: $m = 1/5$

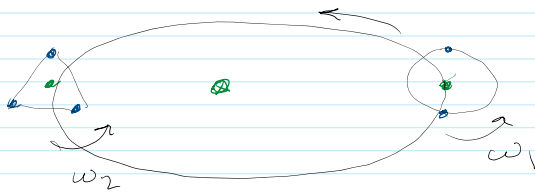
A) circular orbit around mutual COM



Scenario A: Circular mutual orbit ($e=0$)

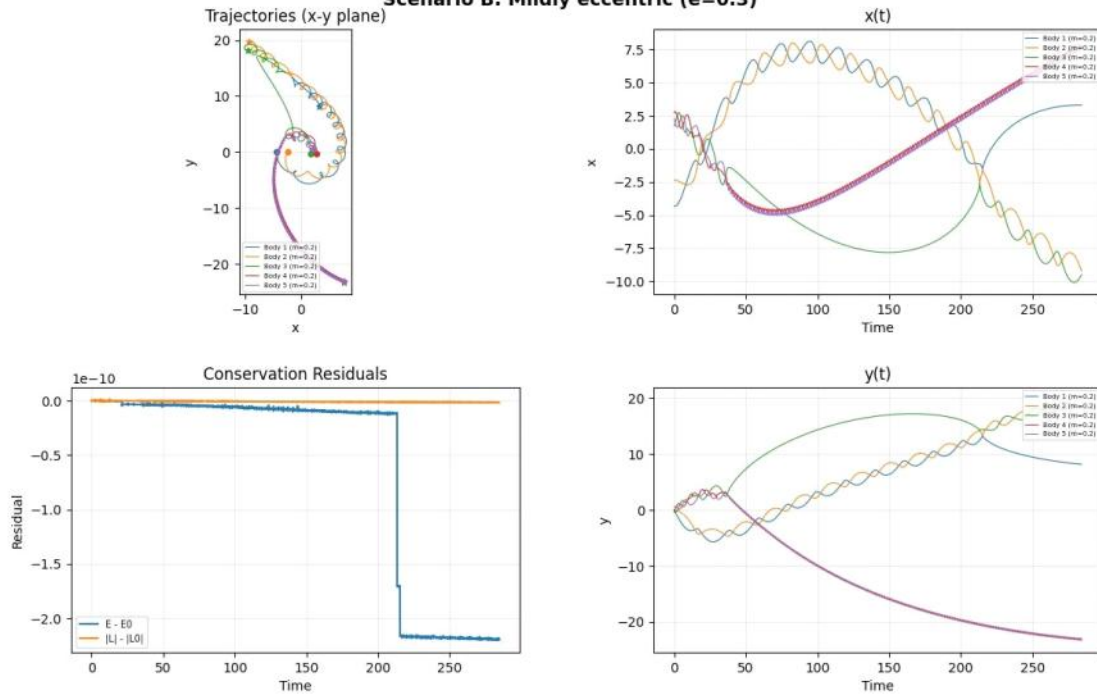


B) elliptical orbit around mutual Com



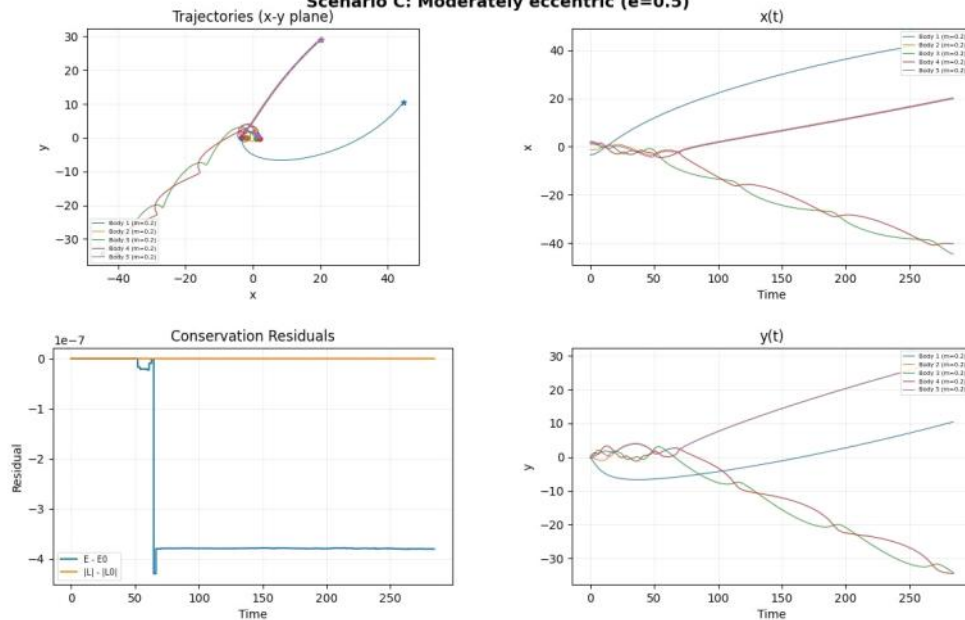
$$e = 0.3$$

Scenario B: Mildly eccentric ($e=0.3$)



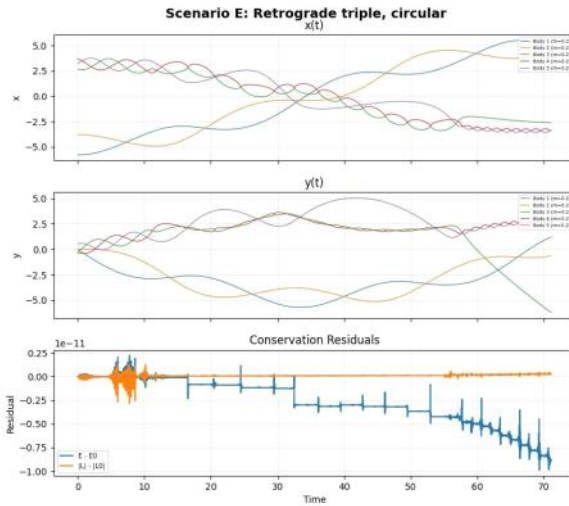
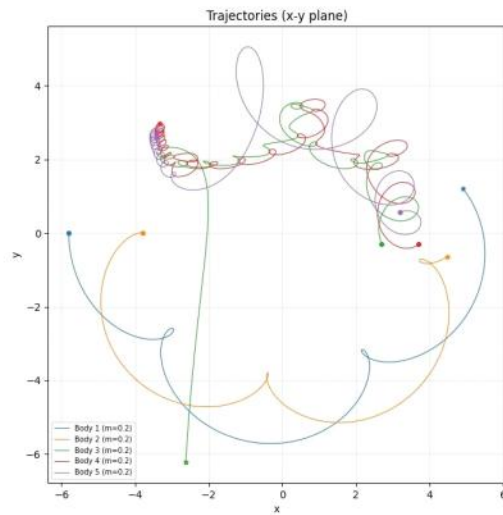
$e=0.5$

Scenario C: Moderately eccentric ($e=0.5$)



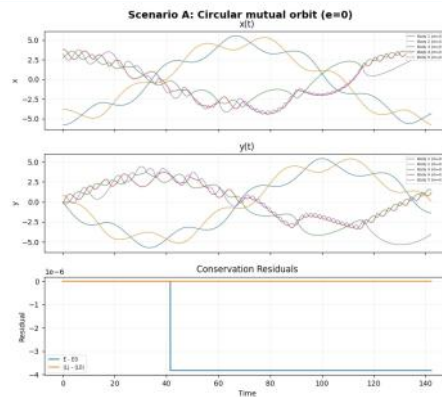
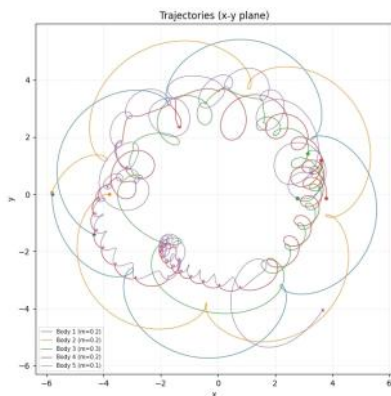
This has 3 body system rotating retrograde to 2 body system, in circular orbit

Scenario E: Retrograde triple, circular



This example shows impact of non-equal masses & their impact on the trajectories of other bodies,

Scenario A: Circular mutual orbit ($e=0$)



In all the scenarios I tested, given enough time at least one body was ejected. The motion is very chaotic & hard to predict. Lighter bodies are more impacted in their orbits when perturbed by other systems.